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SIERRA Lunch & Learn

January 29, 2026

Day 1: Why Sierra

Sierra:

- Is the ABS-approved AI assistant
- Uses best-available AI models
- Resides on ABS IT infrastructure
- Protects client data
- Keeps messages private from AI vendors
- Meets compliance requirements

Other AI Assistants:

- Not explicitly approved for use
- Data leakge

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Day 2: Privacy, Security & Data Governance

Policy Compliance

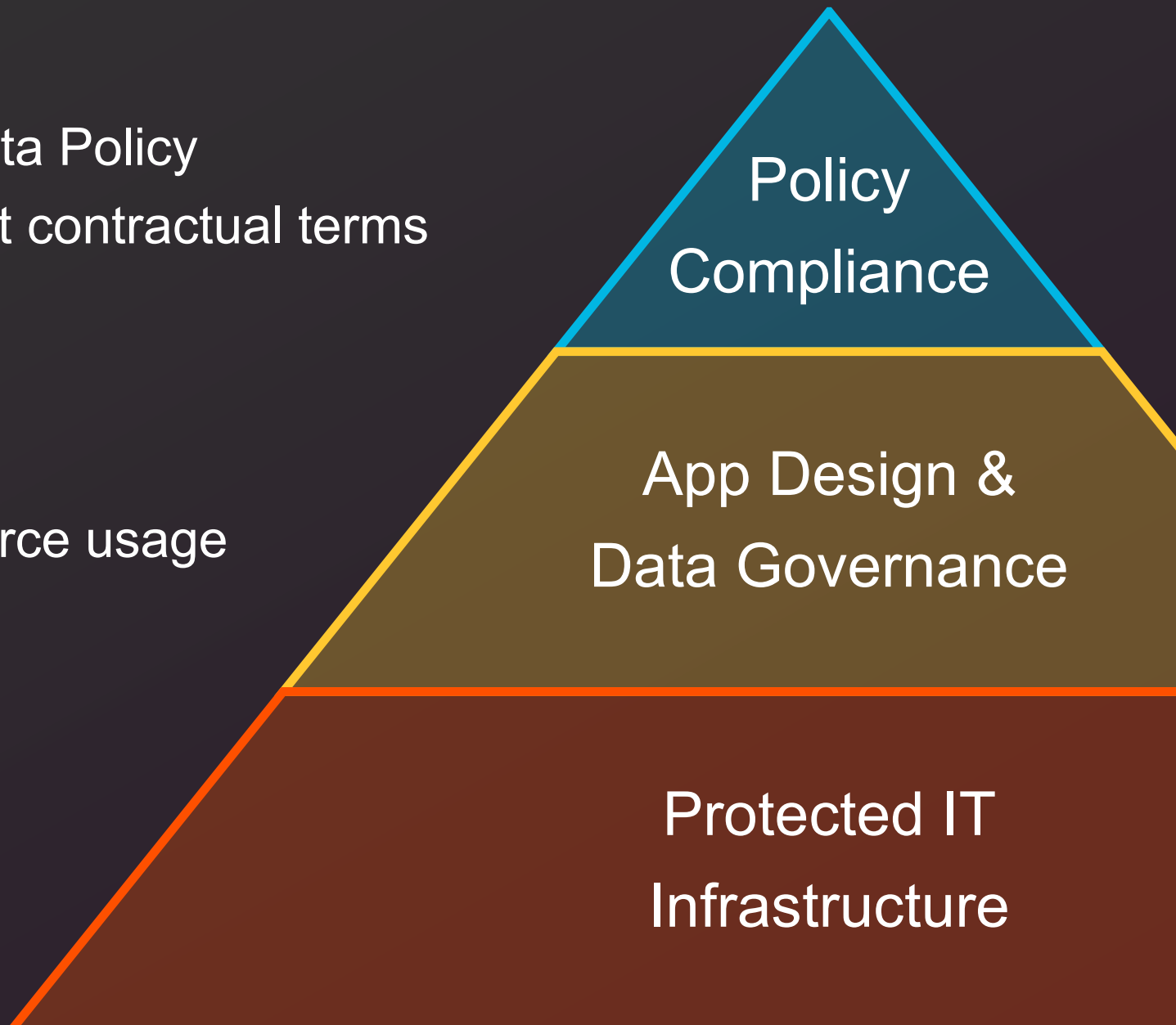
- Users must abide by the Sierra Data Policy
- Data Policy is superseded by client contractual terms

App Design & Data Governance

- Chat threads are private
- Workspaces manage collaboration
- Limited web search/external resource usage

Protected IT Infrastructure

- Approved by ABS cyber teams



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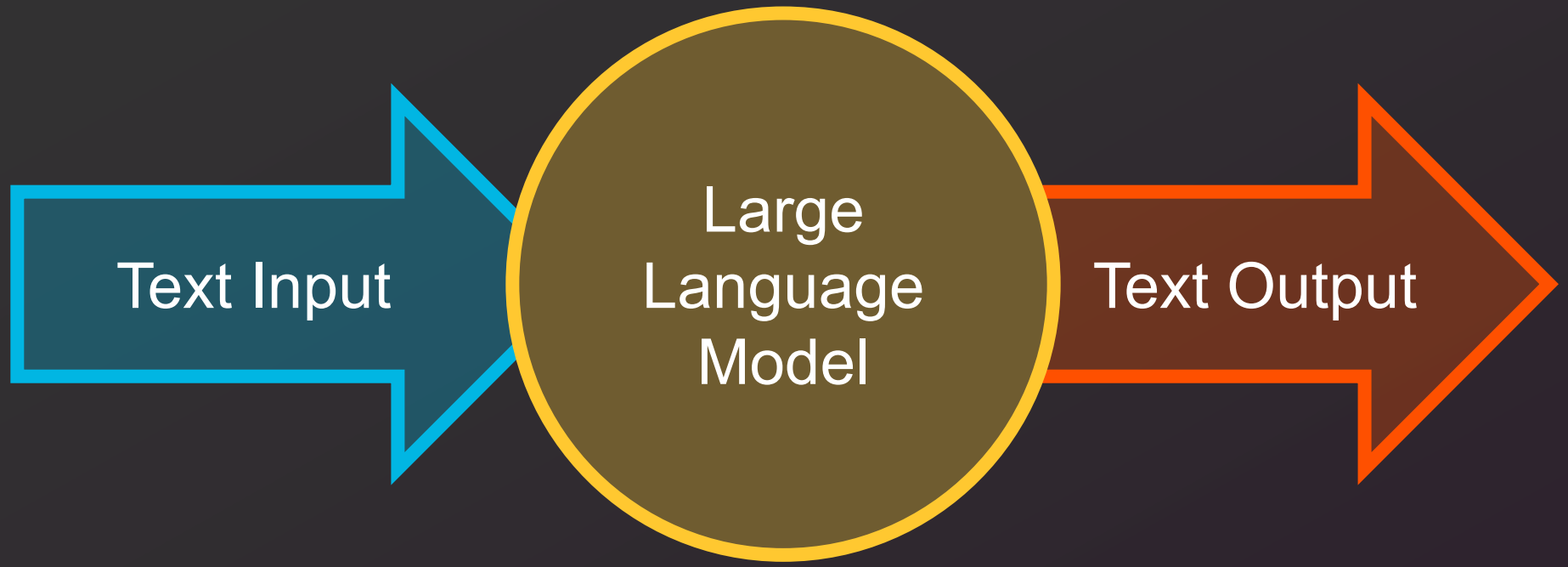
Day 3: What is a Large Language Model (LLM)?



Core Ideas

- An LLM is a calculator that converts a text input (a prompt) to a text output (a completion).
- It stores a large quantity of digital information that it uses in its calculations, giving the appearance of knowledge and logic: Artificial Intelligence.
- The calculator injects randomness to create human-like response

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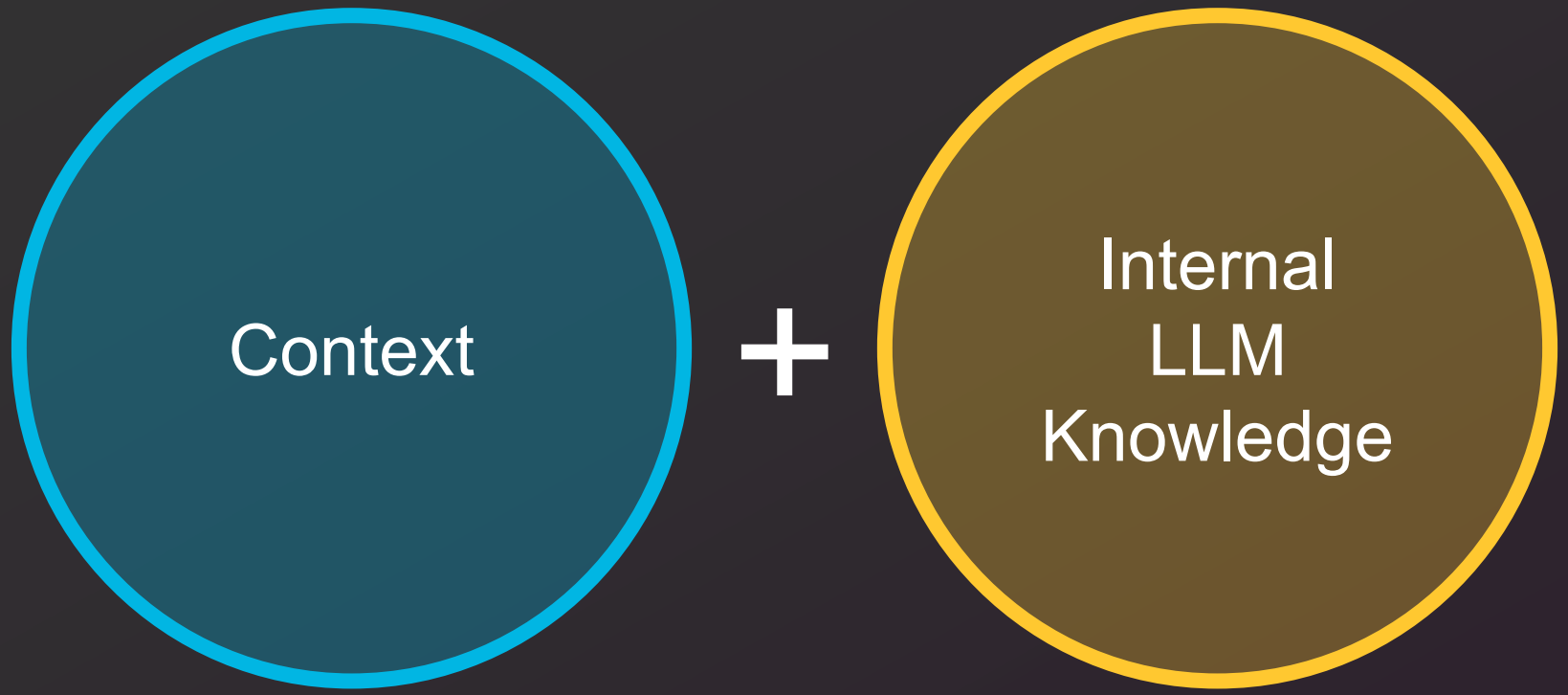
Day 4: What else can LLMs “know”?



Core Ideas

- The LLM has its own general knowledge that “comes with it.”
- It can also react to external knowledge, “Context,” that is provided to it.
- There is **no** other knowledge available to it

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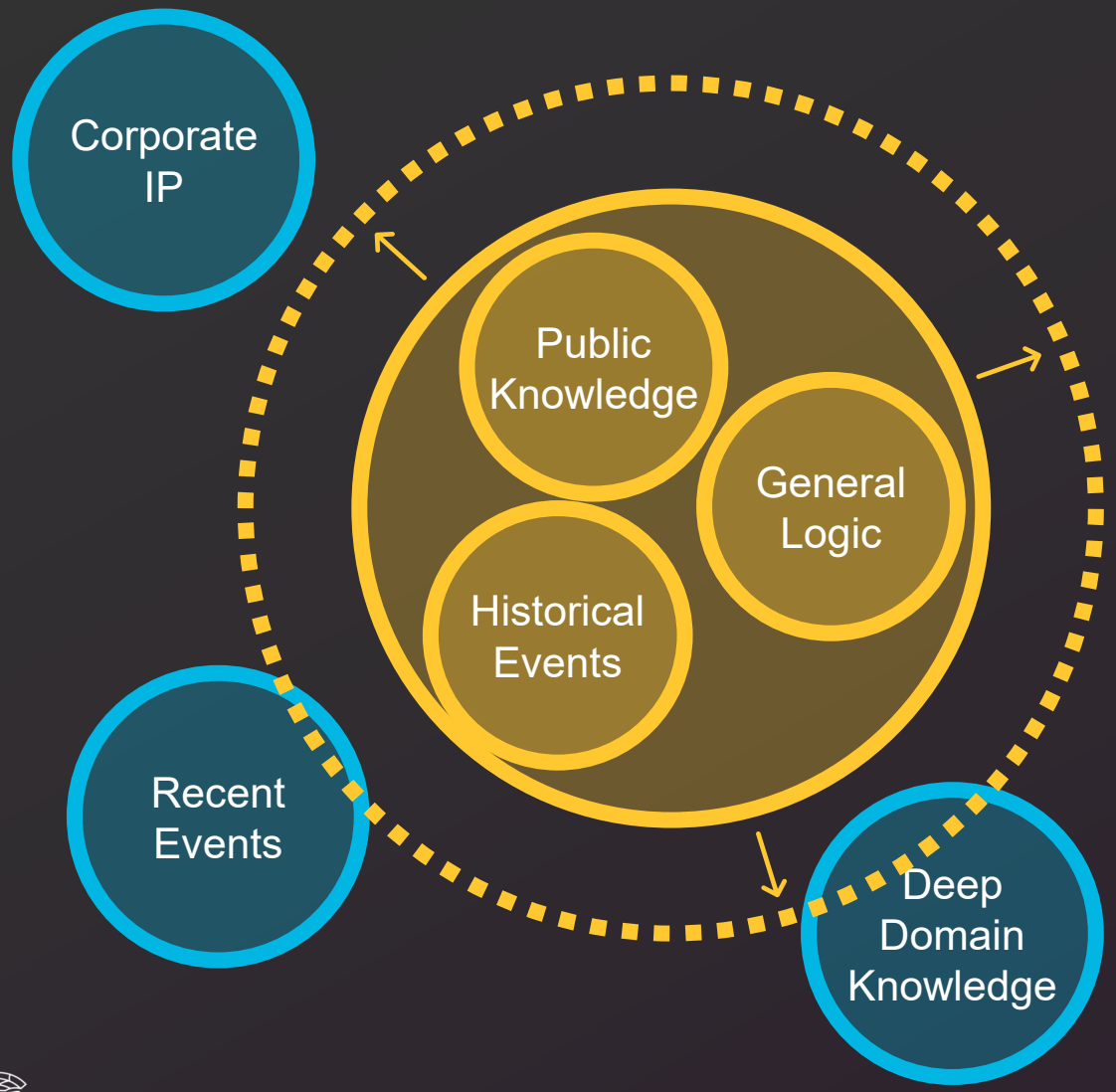
Day 5: What LLMs know and do not know



Internal Knowledge Includes:

- Publicly available knowledge (internet sources)
- Events that occurred prior to the creation of their training data (cutoff date)
- Information & logic that is inferable from statistical patterns in the source text
- *Note: newer LLMs have more recent cutoff dates and deeper domain knowledge*

Other knowledge can be provided as context, but is not "built in"

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Day 6: What is LLM Training?

Core Ideas

- Training is using large amounts of information to calibrate a generative model so that it imitates human intelligence
- LLM's are trained by major AI companies: OpenAI, Anthropic, xAI, Google
- LLMs do not automatically learn from being used.
- ABS does **NOT** train its own LLMs

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Day 7: Tokens & LLM Cost

- Every LLM request incurs a fee
- This fee is based on the number of tokens used as input and generated as output
- The longer your conversation and the more context that is provided, the more expensive the LLM request
- LLMs perform best when “just enough” context is provided and when conversations are not too long
- On average the number of tokens is about 1.3x the number of words

About tokens…

LLMs technically do not process text directly—they process tokens. Tokens are pieces of semantic meaning that can be represented numerically.

LL → [1.2323,0.223,-4.797,3.156,…

Ms → [-0.156,3.583,0.213,-9.397,…

technically → [-4.897,2.196,1.272,…

When you start a fresh topic in Sierra, start a new thread to minimize cost and improve response accuracy.

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Day 8: What is a Hallucination?

Core Ideas

- A “hallucination” is “high probability,” but inaccurate, output from an LLM
- The individual words and phrases make sense, but the answer is wrong.
- This often occurs around the “edges” of the LLM’s training data/knowledge

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Day 9: Basic Prompting

Components of a good prompt

- **Context:** *what background information does the LLM need?*
- **Task:** ***w**hat specifically do you want it to do?*
- **Format:** ***s**pecify the output structure (number of words, bullets, narrative, table, etc.)*
- **Tone:** *how should it sound?*
- **Examples/Constraints:** *what do you need for success?*

Basic prompt tasks

- **Search:** *"What is ____?"*
- **Generation:** *"Write an email saying ____."*
- **Transformation:** *"Take this ____ and make it more ____."*
- **Analysis:** *"Analyze ____ to help me understand ____."*
- **Role-play:** *"Talk to me as if you were a ____ to help me ____."*

Core Idea

Don't overthink prompting. Pretend like you are sending requests to a hyperactive intern who will attempt anything that you ask. Sometimes it won't work, but many times it will.

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Day 10: Advanced Prompting

Speech to Text

IF: you don't know exactly what you want OR you have a lot of material to input

THEN: speech to text can save a lot of time

Prompt Iteration

IF: the initial LLM response is not satisfactory

THEN: provide follow up prompts to trigger Sierra to regenerate an improved result

"First, ask me questions"

IF: you are uncertain of what information Sierra needs to generate an accurate result

THEN: tell it to ask you clarifying questions prior to responding

"Write a better prompt"

IF: you've iteratively improved a request AND you plan to repeat the request in the future

THEN: ask Sierra to generate a better initial prompt for you to use next time

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Day 11: What is Context?

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Intro to AI Center of Excellence

State of AI

- Proactive Agents

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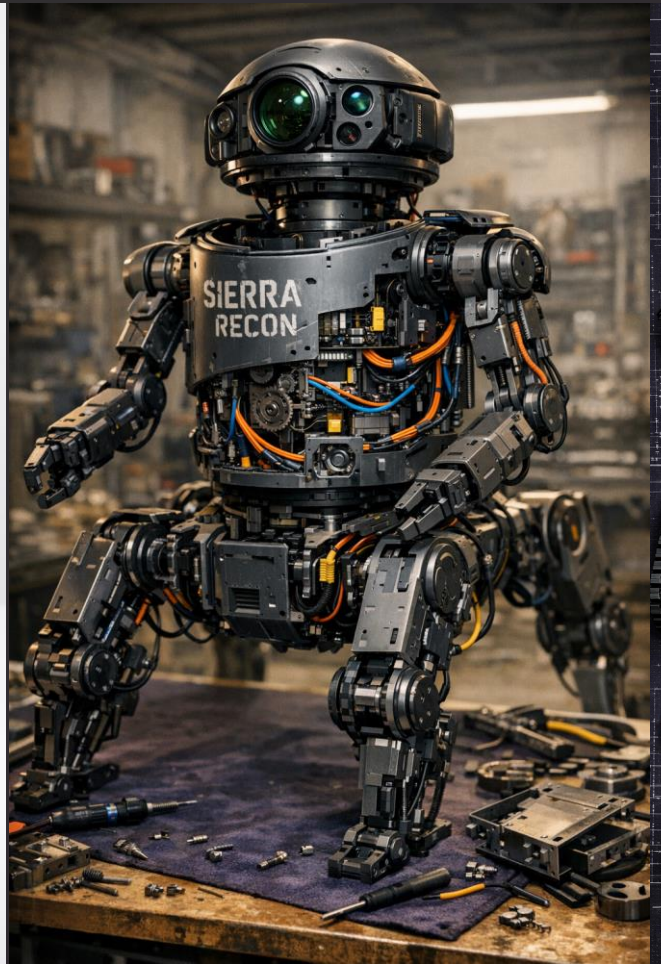
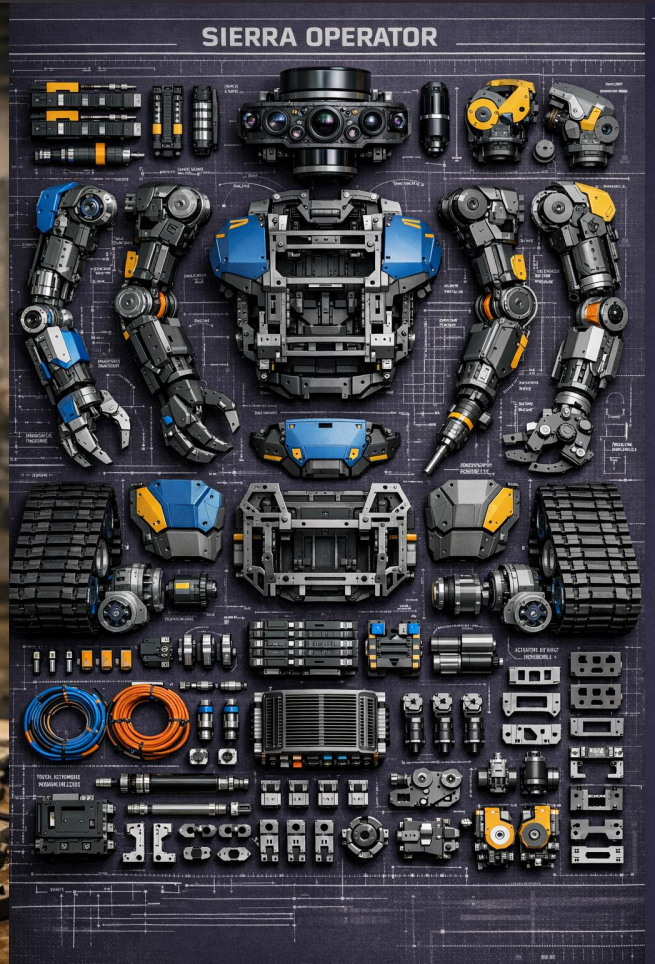
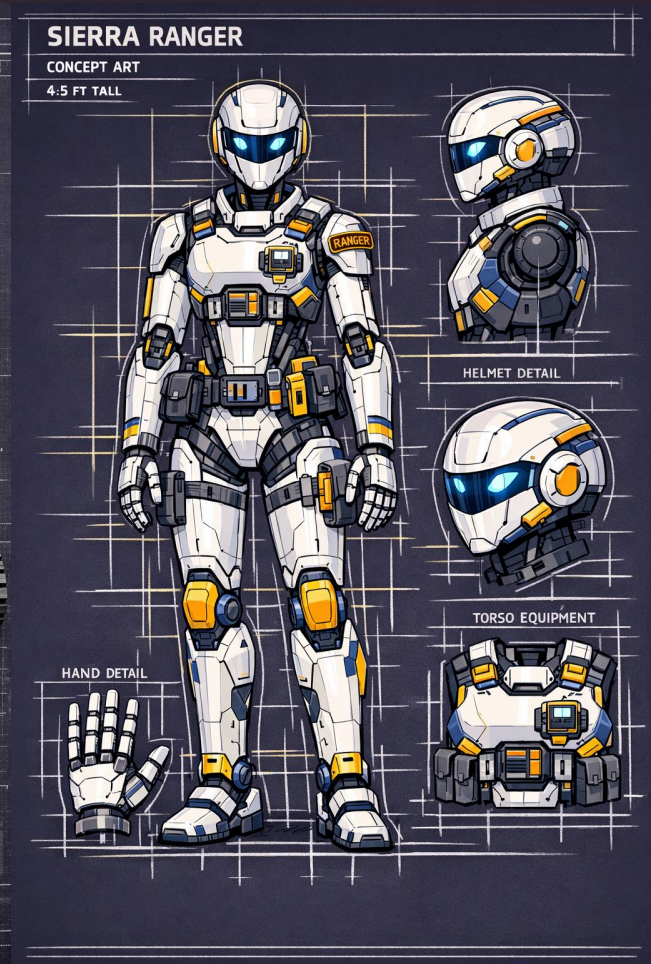
November 2024







Sierra v1 “COMMS” – November 2024

Status:

- Deprecated

Features:

- Chat
- File upload
- Retrieval Augmented Generation (RAG)

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Sierra v2 “RECON” – December 2025

Status:

- Released
- Bug fixes ongoing
- Knowledge base development underway

Features

- Task workspaces
- Personalization
- Web search
- Upgraded tool agent

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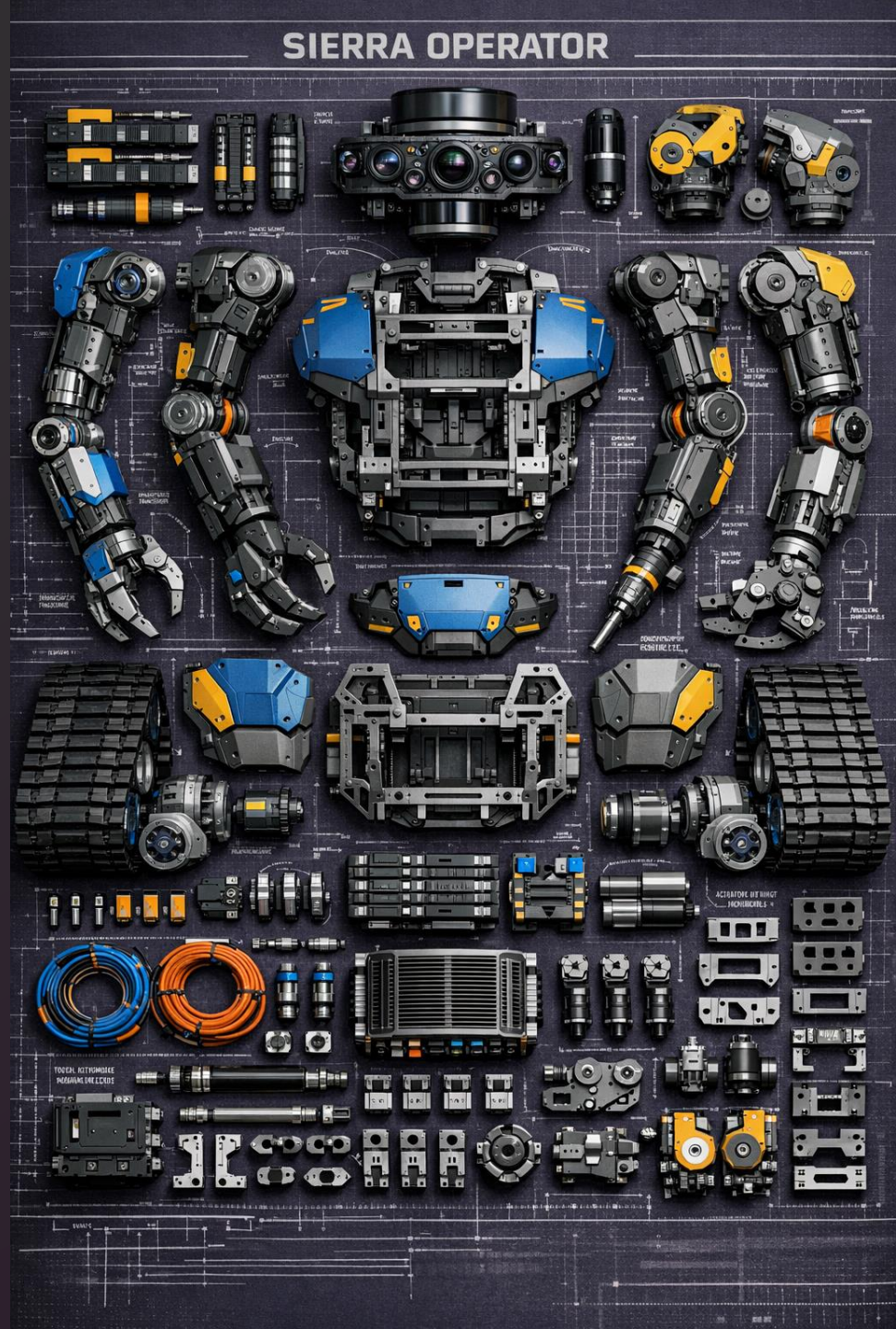
Sierra v3 “OPERATOR” – estimated 2026

Status:

- Gathering requirements
- Prototyping

Features:

- Project workspaces
- Integration with ABS systems (GEMS)
- Agentic file editing (Pad)
- Collaborative AI memory
- Document version control/protection



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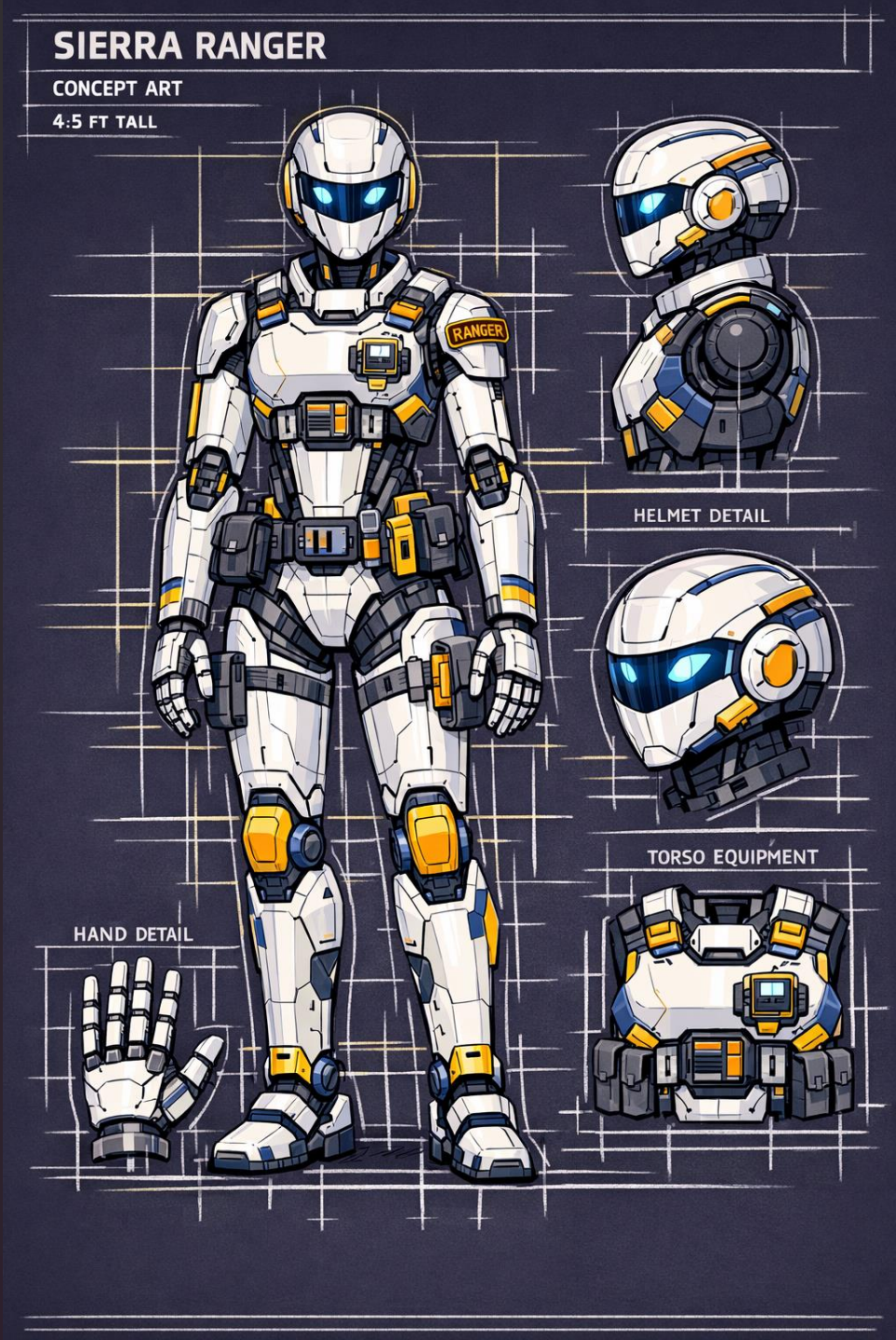
Sierra v4 “RANGER” – estimated 2026

Status:

- Concept development

Features:

- Autonomous agent
- Multichannel communication
- Proactive communication
- Code generation/execution
- Field data ingestion
- Advanced graph memory (CrAInium)



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Planned Tools

- Tango – Document extraction
- Echo – Messaging
- Bravo – Document stamping
- Hotel – AI document version control
- Zulu – Final document formatting

SIERRA utilities:

- Sierra CrAInium – Dynamic graph knowledge base
- Sierra Pad – Document Editing UI
- Sierra API

Key Challenges in Enterprise Deployment

- Unfriendly document types
- Data governance

Important Trends

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- What is our moat:
  - Not just anyone can do class work
  - Consulting side, less of a moat
- Bureau
  - Moat is class services
- Wavesight

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AI COE Scope Areas

1. Governance & Coordination

- Policies & Procedures
- Controls
- Standards & Baselines
- Collaboration
- Communication
- Training

2. Internal Automation

- SIERRA
- PLATO
- Task-specific automations
- Project-specific automations

3. AI-enabled Digital Products

- Predictive ML models in SaaS
- Chatbots w/ ABS repos (External PLATO)

4. AI & Automation Services for Clients

- AI-powered data extraction/cleaning
- AI-powered procedure consolidation
- Chatbots w/ ABS & client repos
- Process automation
- Document intelligence
- Custom AI model development
- Predictive maintenance
- Compliance & risk monitoring

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AI COE Organizational Structure

ENTERPRISE

- Bureau
- Group
- Wavesight

AI Center of Excellence

Requirements →

← AI & Automation Solutions

(Δ) DELTA TEAM
Change Management Specialists

- Build relationships with teams to understand work processes
- Identify opportunities and scope requirements
- Bridge communication between teams and Alpha team on builds
- Train and coach teams on new tool use

(A) ALPHA TEAM
Deep Technical Experts

- Build, enhance & maintain:
  - Cloud infrastructure
  - General AI tools
  - Team-specific AI & automation tools
  - Knowledge bases

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Contact

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Agenda

- Light the fire…
- Sierra feature spotlight
  - User settings
  - Web search
  - File search
  - Dictation
- Use case training
- What’s coming next
- Technical discussion (How do AI memories work?)

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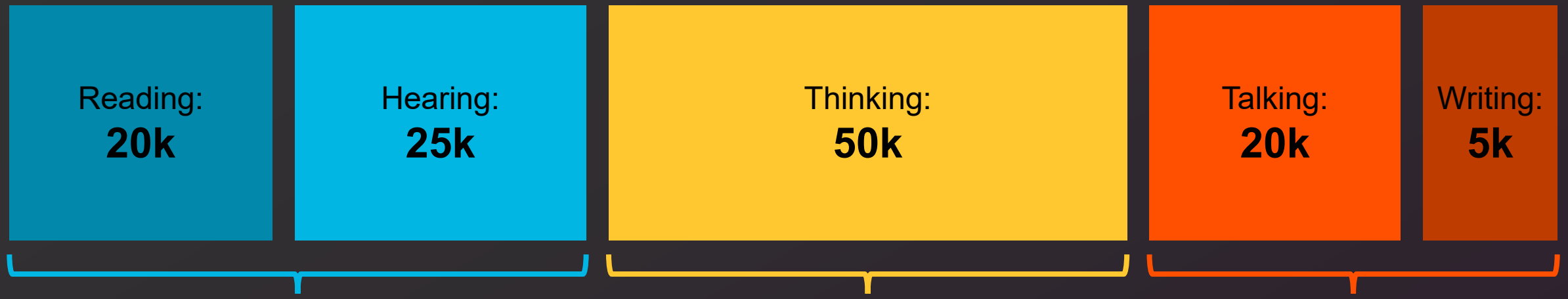
Daily Human Knowledge Work



60k
Input Tokens

60k
Reasoning Tokens

30k
Output Tokens

1 word → 1.3 tokens

$150 of labor

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A Benchmark: Cost per Million Tokens

- We are 50x to 2,000x more expensive than LLMs for processing text

Challenge:
Spend some of your “daily tokens” to figure out how to make AI your personal force multiplier.

	Source	Cost per Million Tokens
	Human	**$1,000**
Frontier	GPT-5.2	$10
Frontier	GPT-5.2 Pro	$120
Frontier	Claude Opus 4.5	$20
Frontier	Claude Sonnet 4.5	$11
Frontier	Gemini 3 Pro	$9
Mid Range	GPT-4o	$7.50
Mid Range	GPT-5 mini	$1.50
Mid Range	Claude Haiku 4.5	$3.50
Mid Range	Gemini 2.5 Flash	$1.50
Budget	GPT-4o mini	$0.50
Budget	GPT-5 nano	$0.30
Budget	Gemini 2.0 Flash	$0.30
Budget	Gemini 2.5 Flash Lite	$0.30
	AI	**$0.5 to $20**

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Sierra's Research

CONFIDENCE LEVEL BY METRIC

Metric	Confidence	Reasoning
Words spoken	⭐⭐⭐⭐ High	Multiple converging studies; meeting data robust
Words written	⭐⭐⭐⭐ High	Email tracking data is precise and consistent
Words read	⭐⭐⭐⭐ High	Email + document data well-documented
Words heard	⭐⭐⭐ Moderate	Listening research consistent but harder to measure
Words thought	⭐⭐ Low	Highly speculative; no direct measurement possible
Salary data	⭐⭐⭐⭐⭐ Very High	Official BLS government statistics
AI pricing	⭐⭐⭐⭐⭐ Very High	Official provider pricing as of Jan 2026

This analysis synthesized data from 50+ academic papers, government statistics, industry reports, and official pricing sources, focusing on 2024-2026 data where available.

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The AI Disadvantage

- Motivation
- Memory
- Specialized Expertise
- Visual Processing
- Agency

Demo

Key Techniques

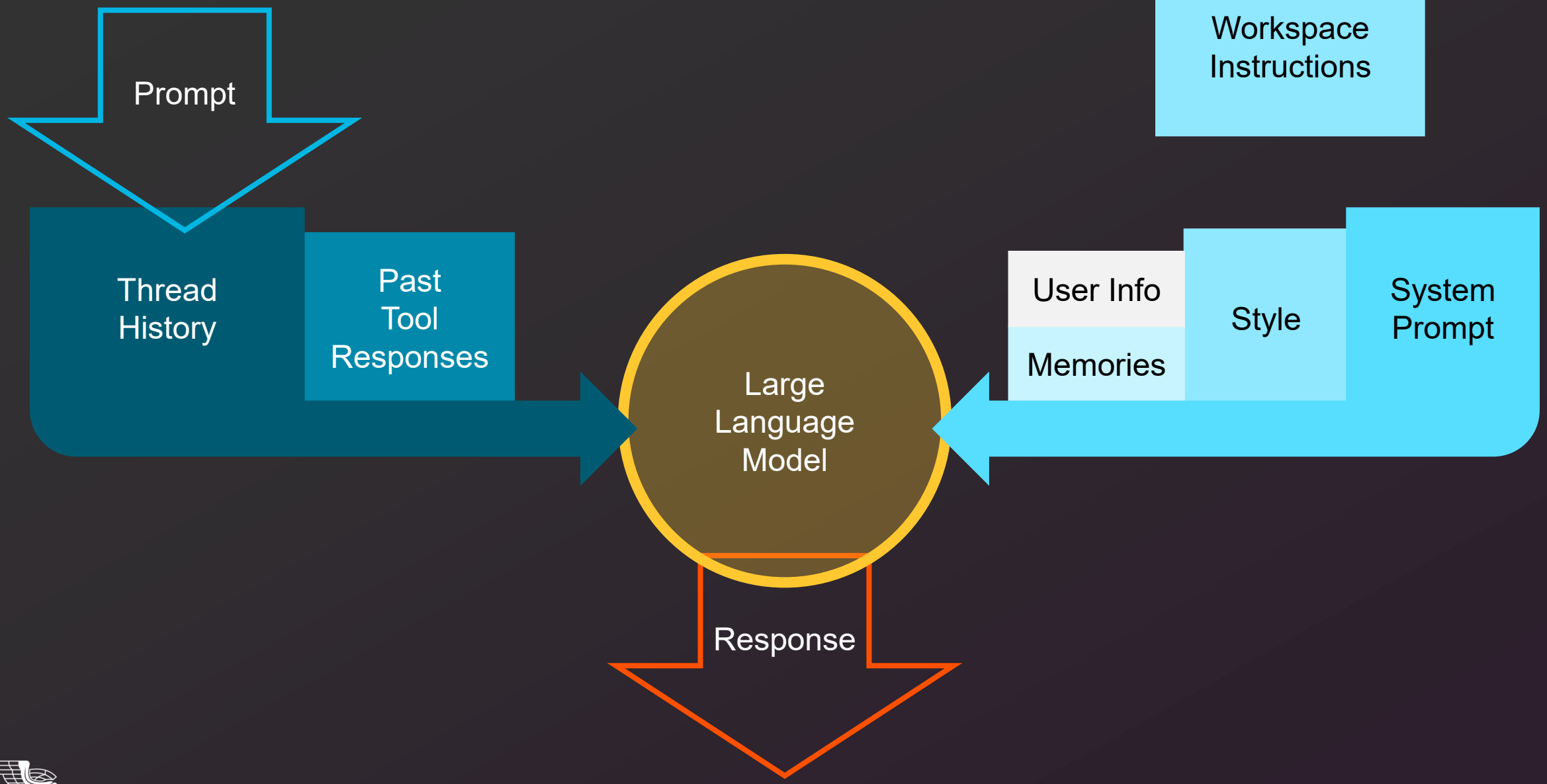
- Dictation
- Web search
- “Summarize my instructions as a new prompt”
- Use workspaces for frequent tasks
- Refresh is your friend

What’s Coming Next?

- Sierra app stability – resolve scaling issues
- File handling
  - Documents Service stability/enhancements
  - Extraction options
  - More supported file types (especially Excel)
  - File re-upload feature
  - Extracted content viewer
- Workspace enhancements
  - GEMS workspaces
  - Workspace memories
  - Workspace agents
  - Workspace templates

Technical Comments: Context Engineering

Context Engineering



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Project Workspace

Task-Specific Workspace (Agent)

Task-Specific Workspace (Agent)

Project Workspace

Transcript Summary 1

Proposal 1

Transcript Summary 2

Proposal 2

Transcript Summary 3

Proposal 3

Project Workspace

Transcript Summary 4

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One-on-One Feedback Summary

- Conducted 18 one-on-one sessions across consulting, cyber, MI, engineering, marketing, and operations
- Captured 50+ app ideas, 75+ feature requests, 30+ knowledge base integrations
- Clear patterns emerged across multiple business units
- Strong enthusiasm + clear adoption blockers identified

What Users Want us to Build

Project-Centric Workspaces (8 mentions)

- Auto-provision workspaces when GEMS numbers created
- Auto-invite assigned team members
- Acts as project container for documents, standards, automations

Proposal Generation & Repository (7 mentions)

- Search past proposals by work type, client, site
- Auto-generate new proposals in client style

Meeting Transcription & Note-Taking (5 mentions)

- Teams/Meet integration for internal meetings
- Field assessment audio capture with auto-transcription

What's Preventing Full Adoption

Word/Excel Formatting (8 mentions)

- Manual reformatting costs 20+ minutes per document
- **Cost:** $100K+ per large project (2,500 docs example)
- **Quote:** *"Sierra is supposed to make us more competitive, but with the formatting problem, we're actually less competitive."*

Standards Integration (10 mentions)

- Users manually downloading from Intertek/I2I for each project
- No automated updates; no cross-standard querying
- **Opportunity:** We already have enterprise licenses

Workspace-Level Memories (7 mentions)

- Teams need shared context within project workspaces
- Current: Each thread and person is siloed
- Blocking team collaboration workflows

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Brainstorming Concepts for Project Workspaces

- Workspace Memories
- Shared Folder/File directory - Rab
- Shared File Creation (.md) - Rab
- Project Templates (PSM, Facility Siting, etc.)
- Loadable Workspace Agents
- Tango Documents

Lots of files, multiple people working on it.

Batch –

Availability

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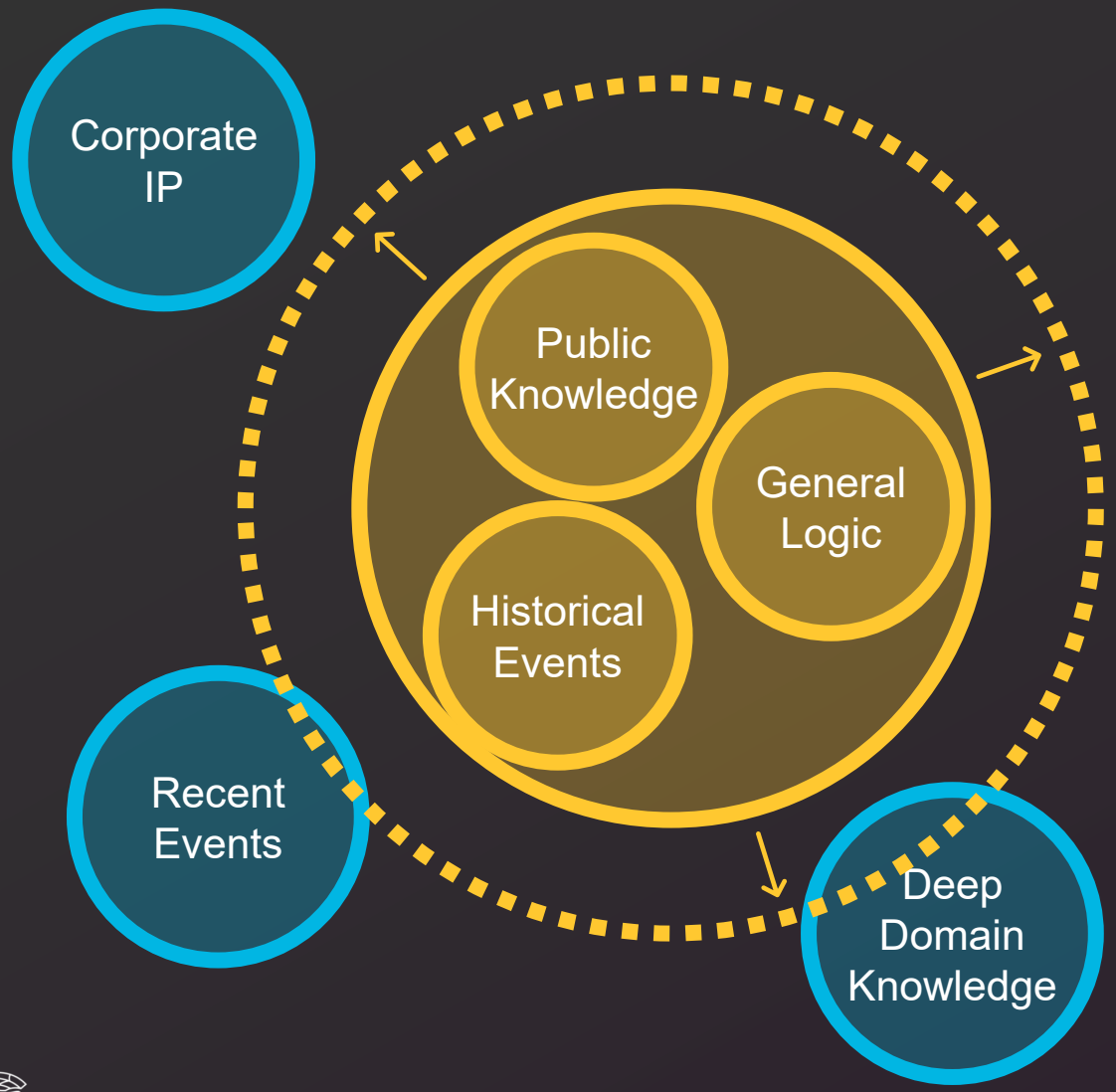
Day 5: What LLMs know and do not know



Internal Knowledge Includes:

- Publicly available knowledge (internet sources)
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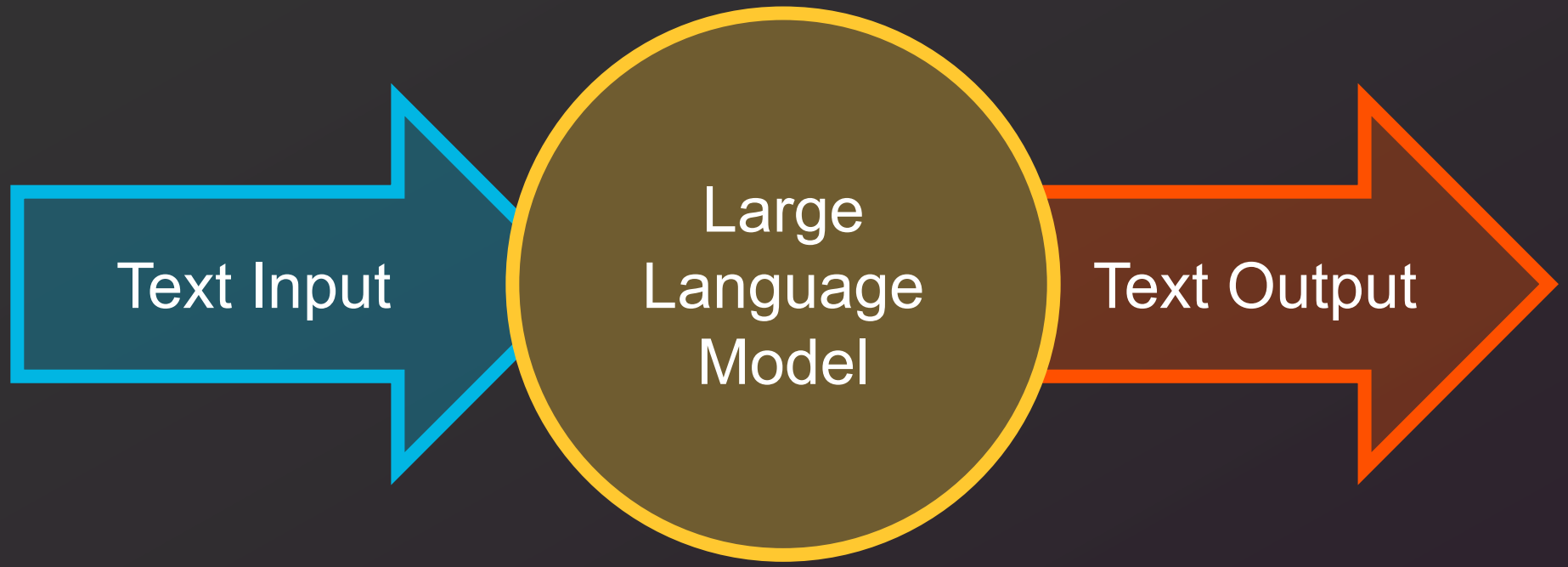

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Day 3

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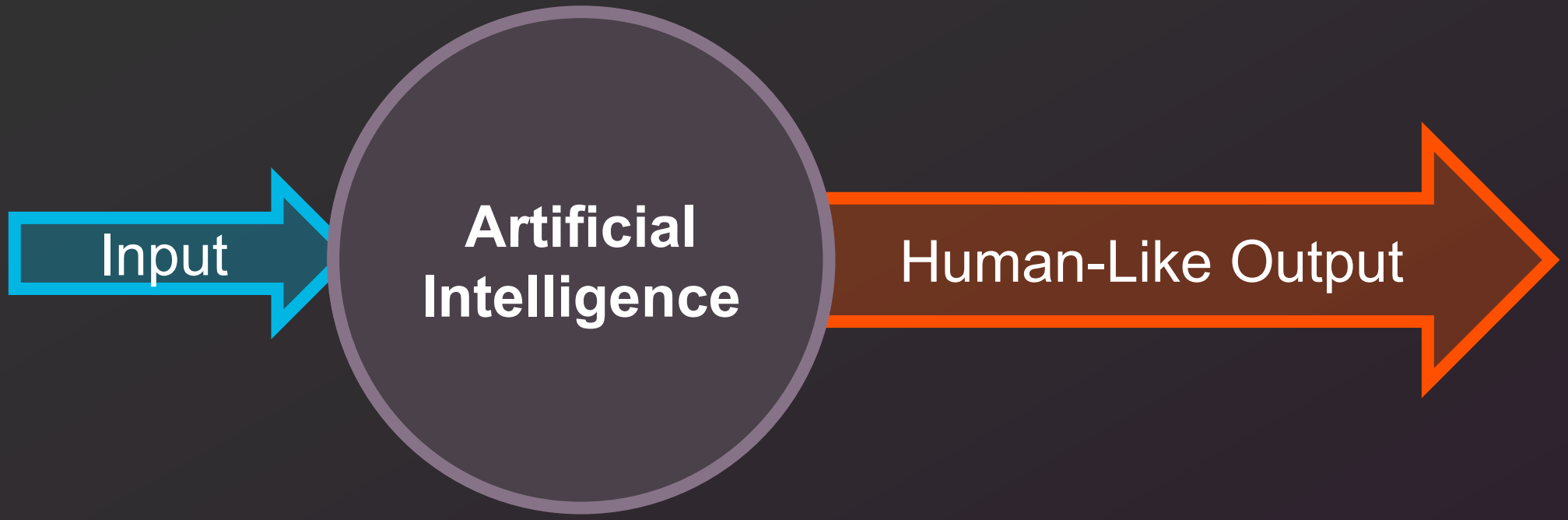
What is Artificial Intelligence?



Core Ideas

- A human-like response to a task/stimulus.
- Machines performing tasks that typically require human intelligence.

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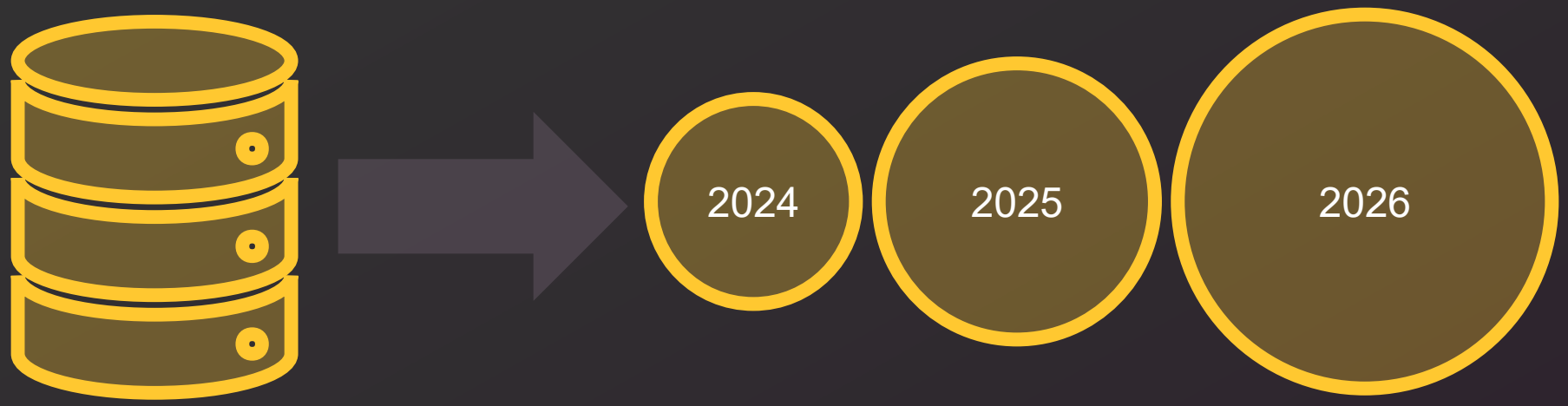
What is LLM Training?



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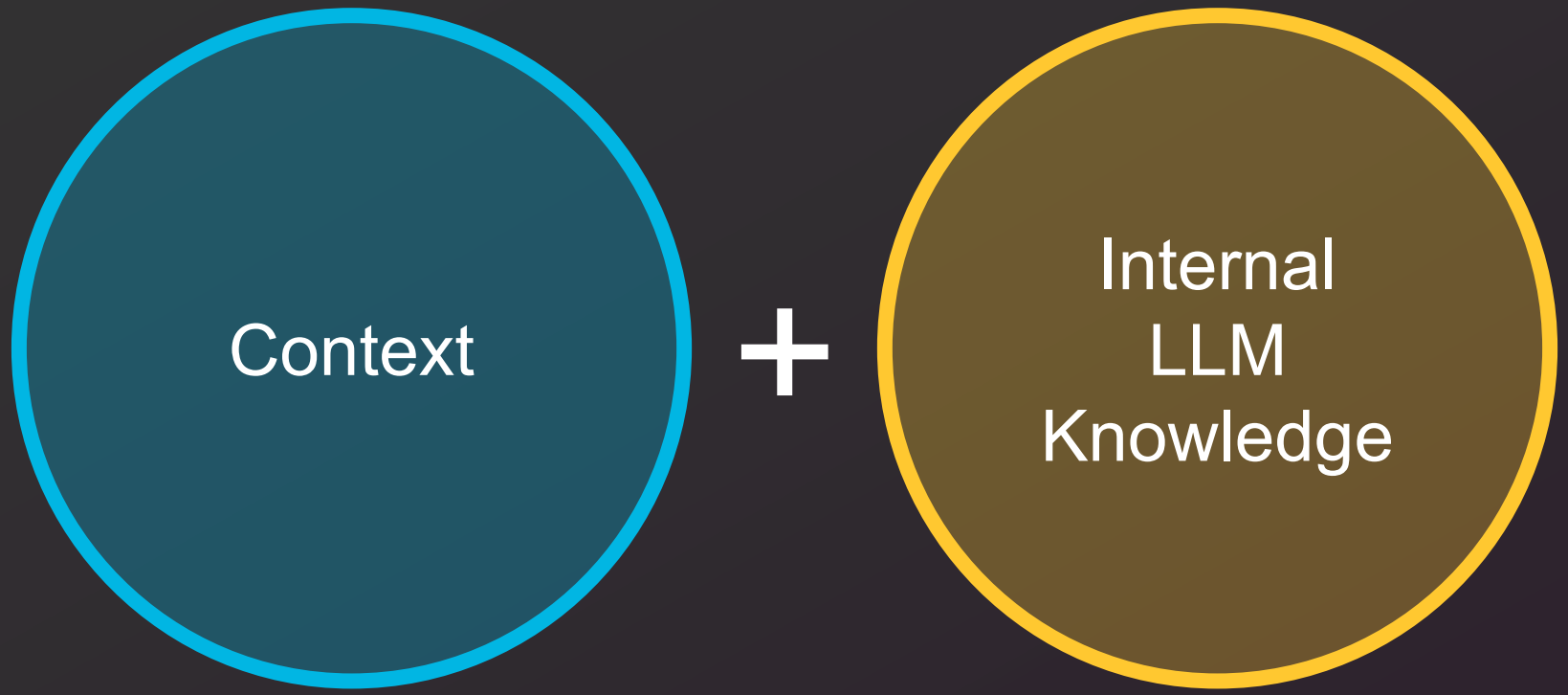

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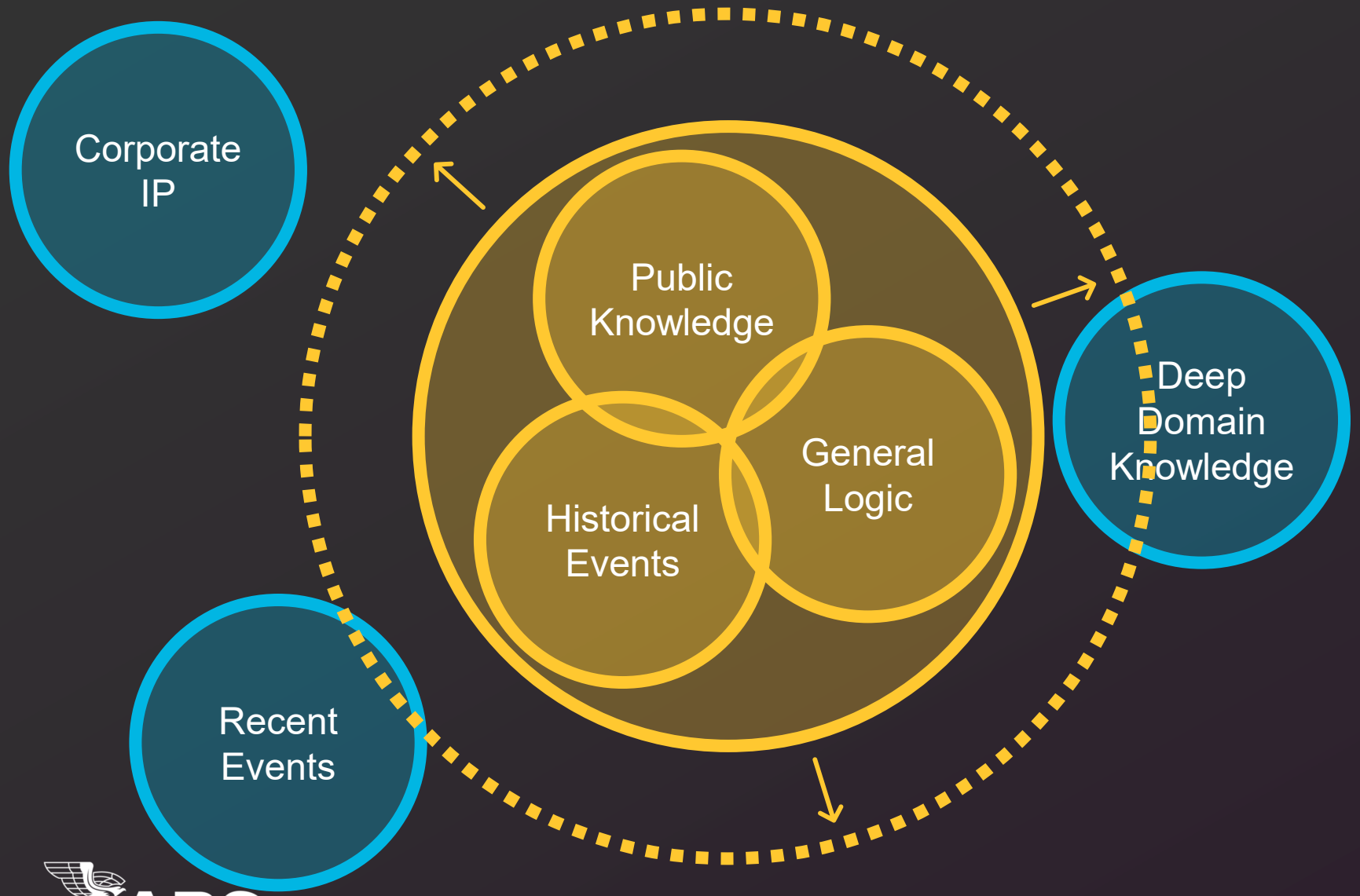
What kinds of knowledge are “internal” to an LLM?



Core Ideas

- Therefore, they kno
  - publicly available
  - Events that occur the creation of th data
  - Logic that is infe these sou
- There internal knowl not include: Corporate History, Deep Tec Knowledge
- AI providers are co expanding the limit knowledge

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What is a Hallucination?

Core Ideas

- A “hallucination” is “high probability,” but inaccurate, output from an LLM
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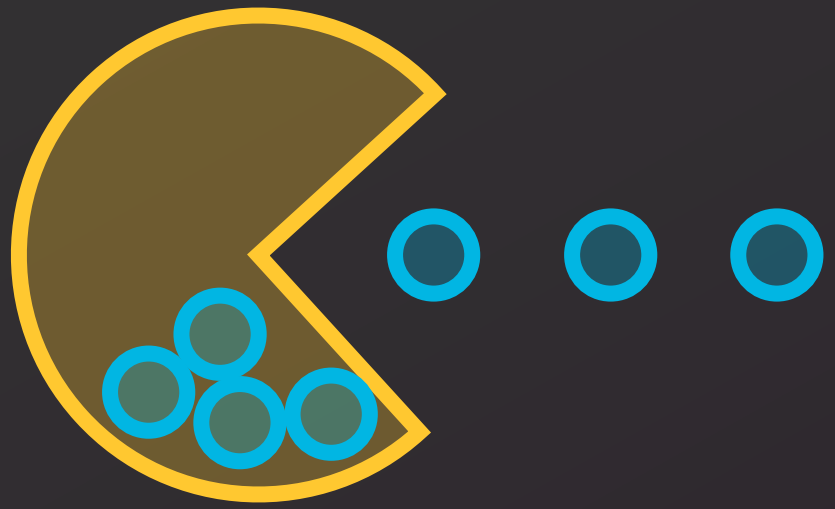
What is fine-tuning?



Core Ideas

- Fine-tuning is doing additional training on an already-trained LLM
  - It usually diminishes the general accuracy of the LLM
- It can improve an LLM's knowledge of specific information or styles
  - It is a way to convert context into "internal knowledge"
- Think of it like brainwashing… the brainwashed LLM may learn to respond properly to certain questions, but becomes unstable in other areas.
- By providing better context, you can generally produce the safe effects as fine-tuning, without the negative side-effects.

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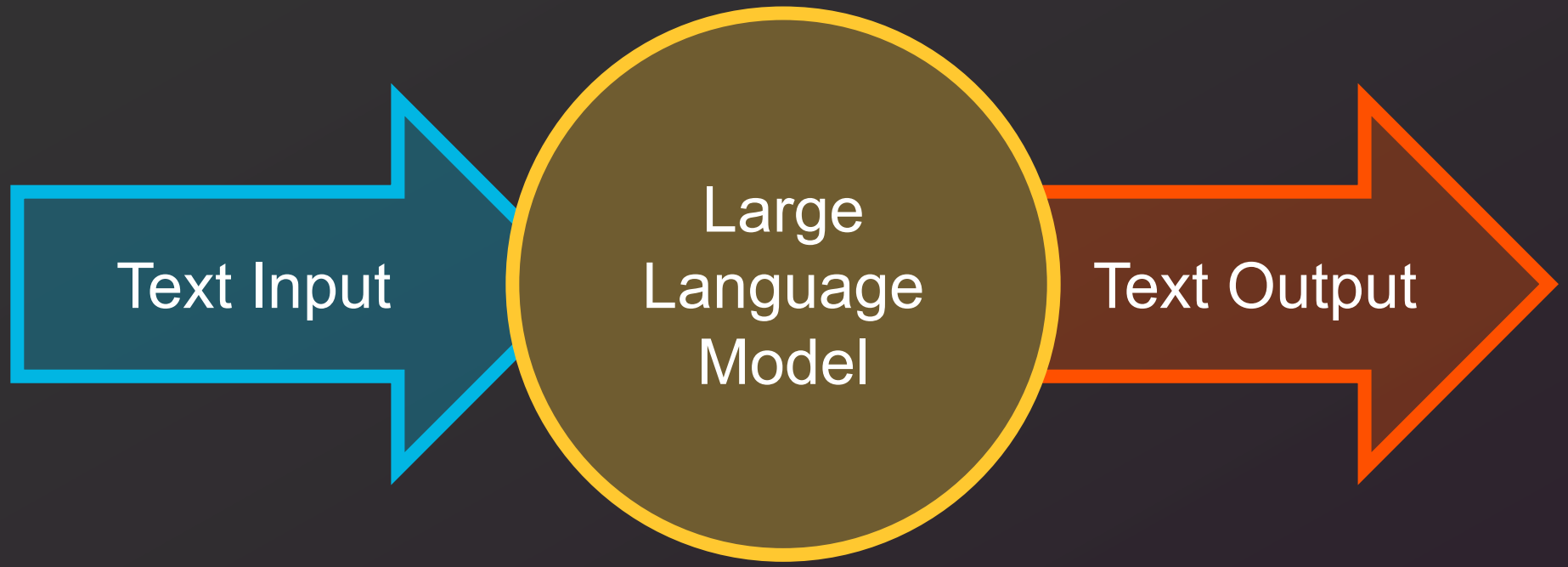
What is a prompt engineering?



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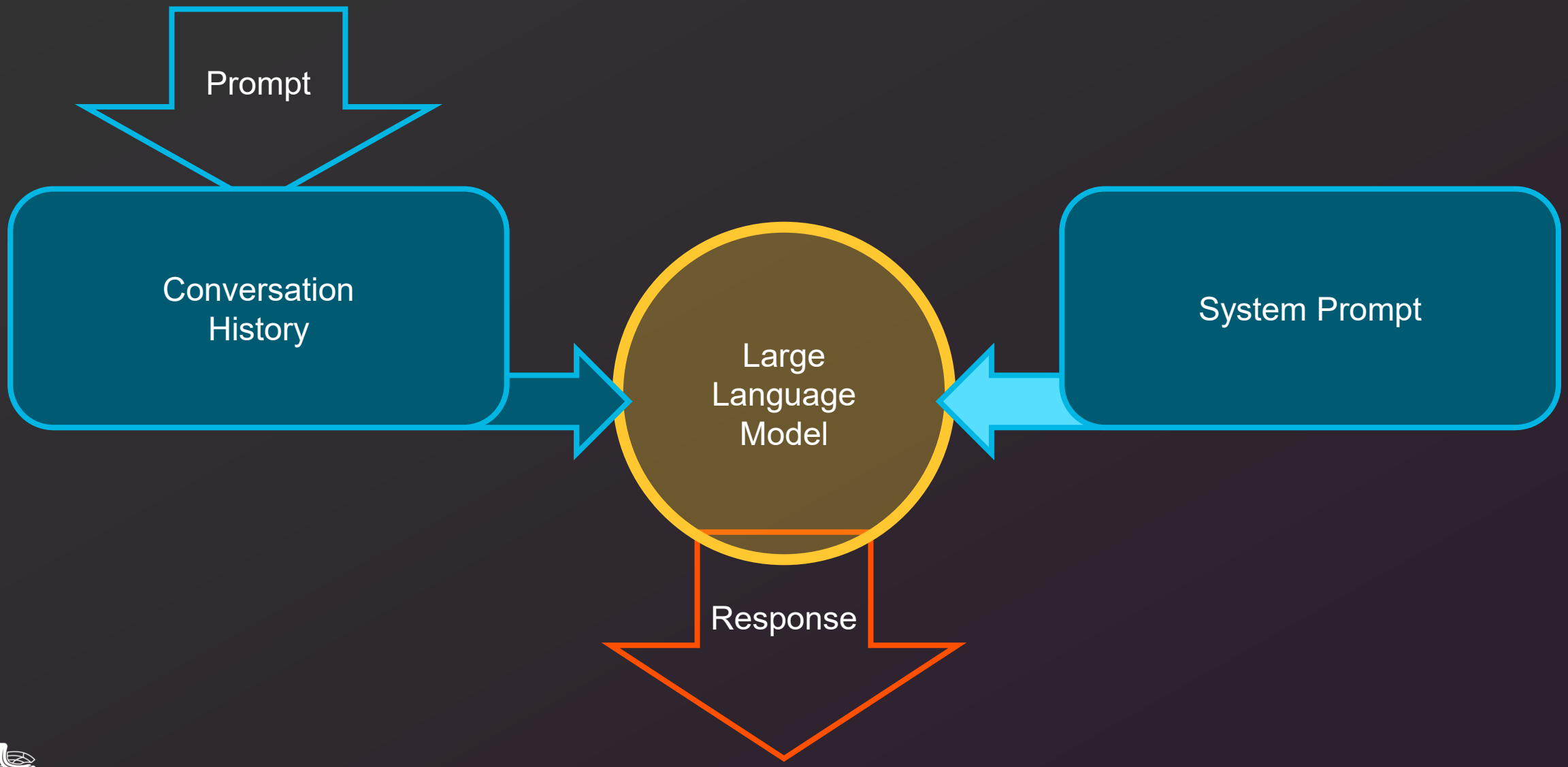
What inputs do LLMs take?



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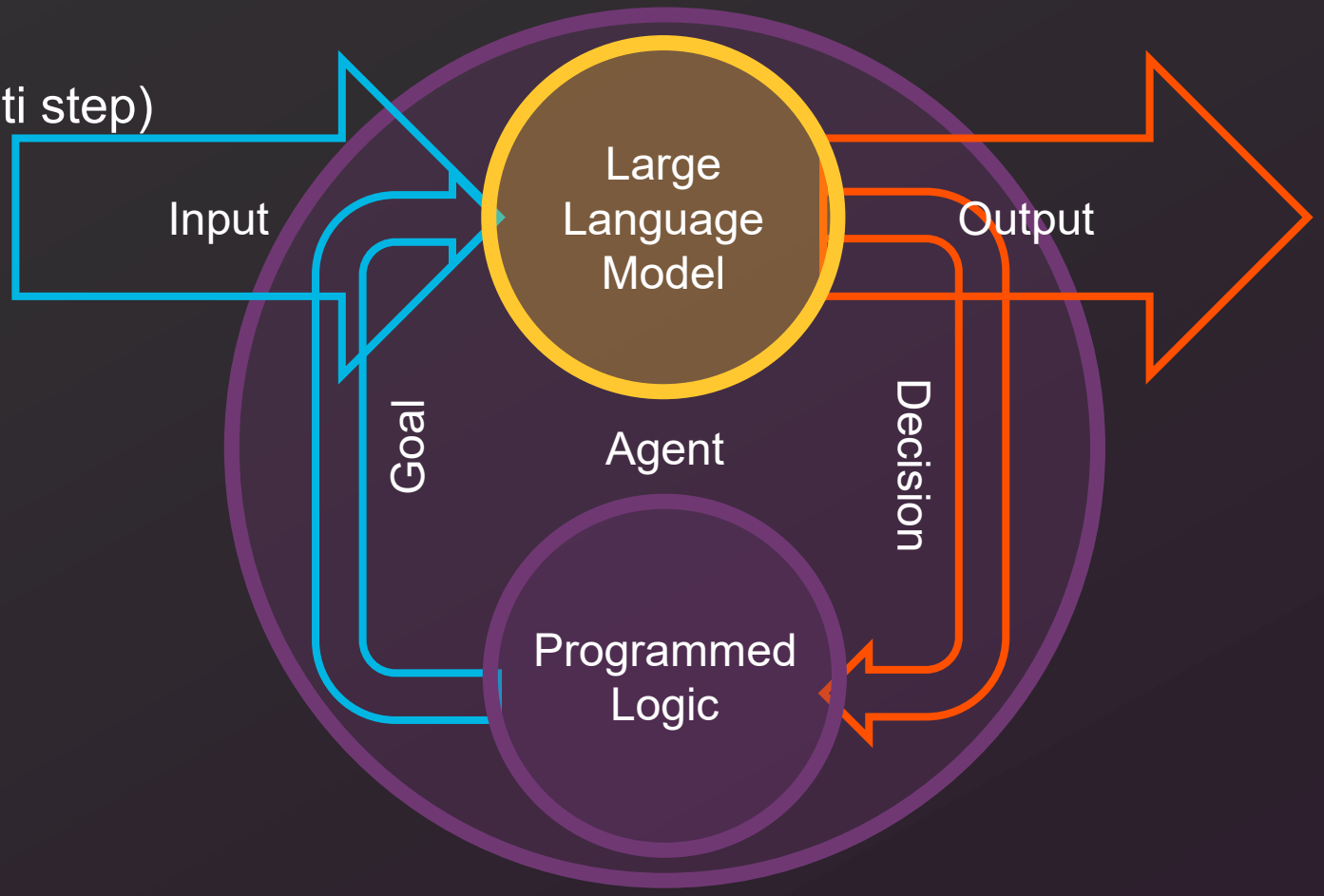
What is a system prompt?

What is an agent?

- An agent:
  - Receives inputs
  - Uses tools
  - Makes decisions
  - Pursues goals (multi step)



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What is an AI “Tool”

Using the jargon that we’ve shared: a tools is a way for an AI to either:

- Retrieve context
- Use LLM output to take an action

How does an agent work?

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What is an AI tool?

Using the jargon that we've shared: a tools is a way for an AI to either:

- Retrieve context
- Turn output into action

- Questions:

How to save? Propose to store as objects on the workspace

options for types, metadata

How to retrieve?

When to save?

* Key word: “remember”

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